04.11.2023

Computer Memory: RAM

Exercice 01:

1) Bus d'adress $= \log_2 4096$

$= \log_2 2^{12}$

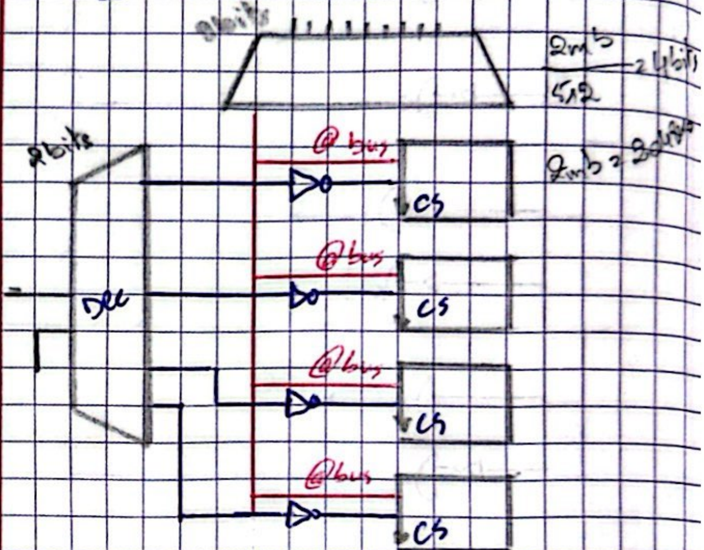
$= 12$ bits

2) Bus des donnée = 32 bits

memory size = size of dataBus $\times 2^{\text{size Adress bus}}$
(size of word) (Num of words)

Exercice 02:

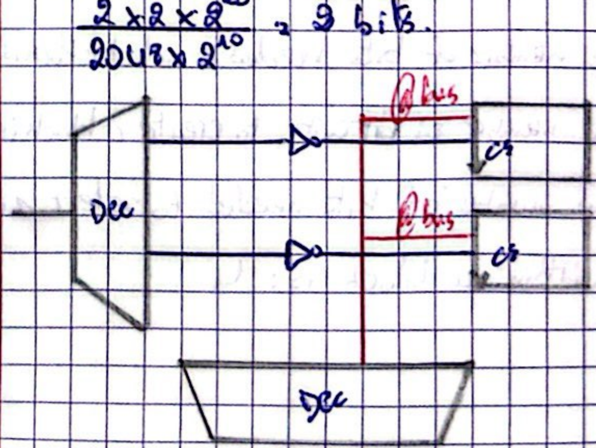
case 1: 512 K. words of 8 bits



Case 2: 2048 kib words of 4 bits.

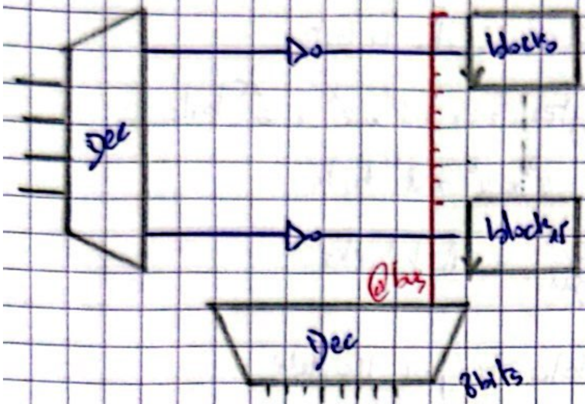
so we have 2 blocks of memory.

$\frac{2\times 2\times 2^{20}}{2048\times 2^{10}} = 2$ bits.



Cas 3: 256 kib words of 4 bits

$\frac{2mb}{128} = 16$ block



cas supl: 512 kib words of 16 bits

size of block $= 2^9 \times 2^{10} \times 2^4 = 2^{23}$

number of necessary blocks $= \frac{2^{24}}{2^{23}} = 2$

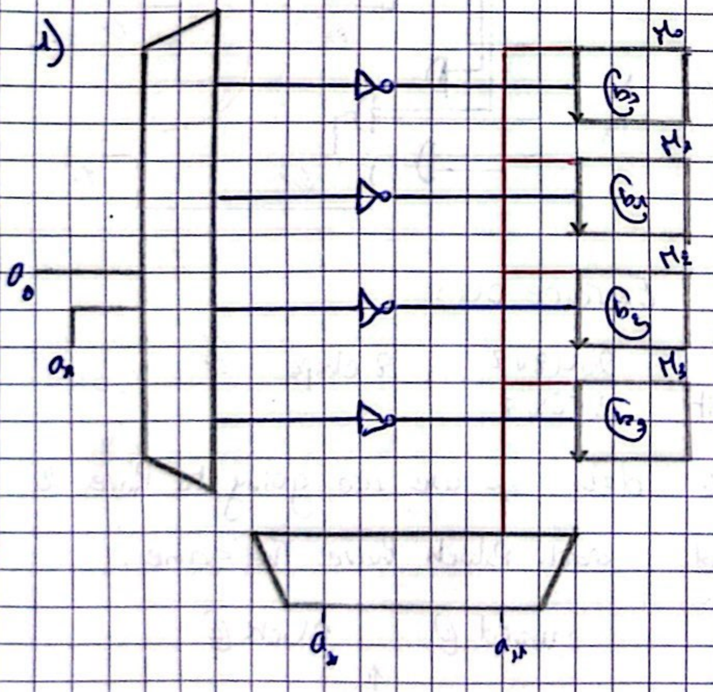
size of Data bus = 16

size of adress bus = 20

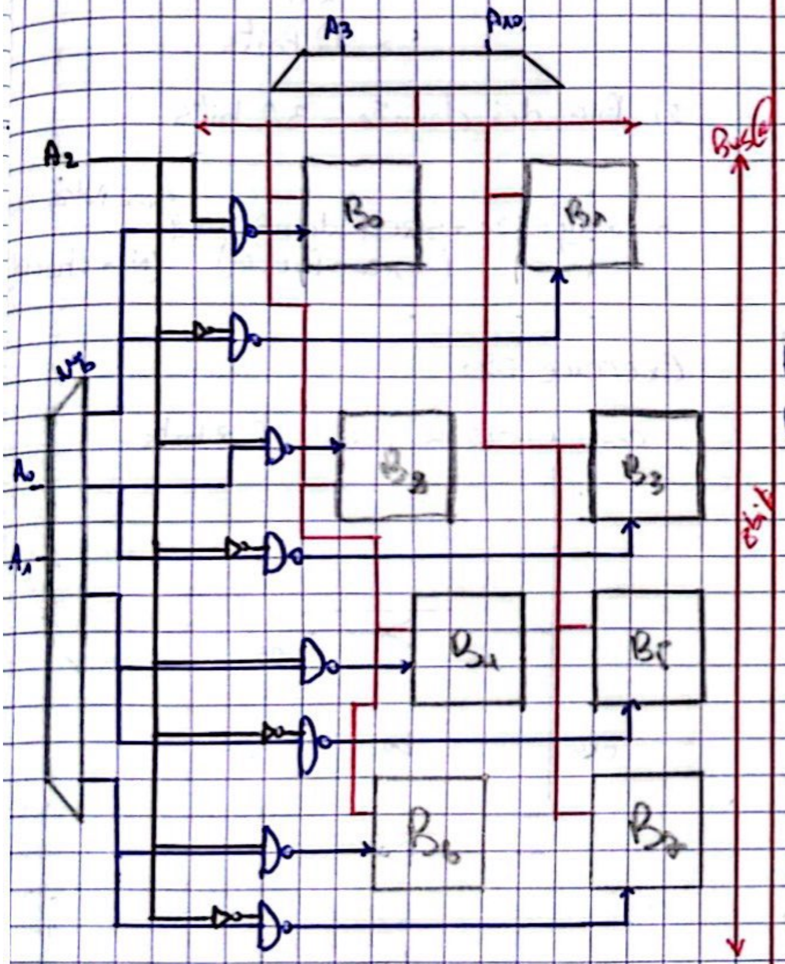
number of words $= \frac{2^{24}}{2^4} = 2^{20}$

Exercice 03:

1)



N°b	A_3	B_0	B_1
0	0	1	1
0	1	1	1
1	0	1	0
1	1	0	1



Exercice 05 :

- The address buse size is : 16 bits
- The nember of bits to select a module is: 1bit
- The nember of bits needed to select a block is: 1bit
- The number of circuits to create a block is: 2
- The number of bits needed to select a word within a block is: 4

2) tailles de la mémoire 4KBytes :

. 4 modules.

Block :

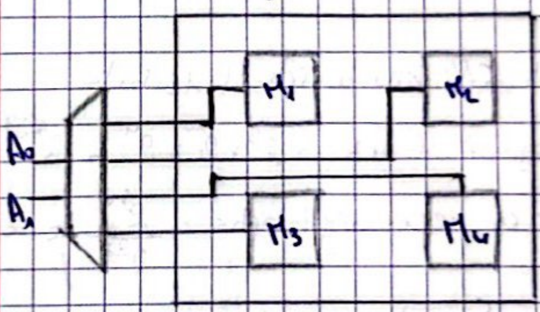
→ 512 mots de 4 bits.

$4kbytes = 4 \times 2^{10} \times 2^3 = 2^{15}$ bits.

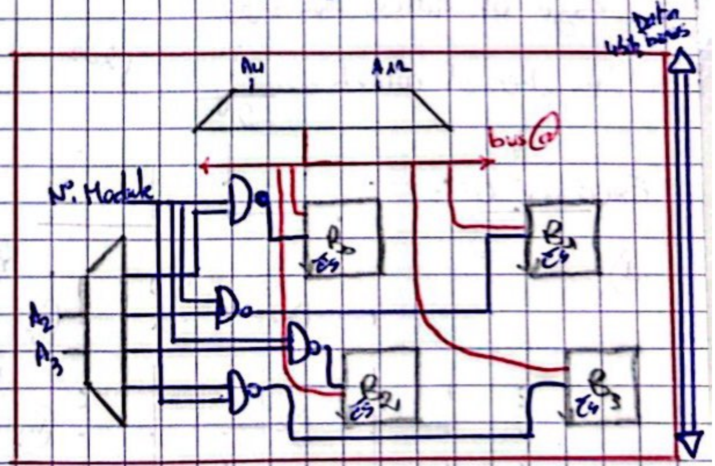
$2^9 \times 2^2 = 2^{11}$ bits.

$= \frac{2^{15}}{2^{11}} = 16$ blocks.

4 blocks par Module :



bus address size = 15



Exercice 04 :

$\frac{2048 \times 8}{256 \times 8} = 8$ chips.

d=4 => we are going to have 2 each Block have The same.

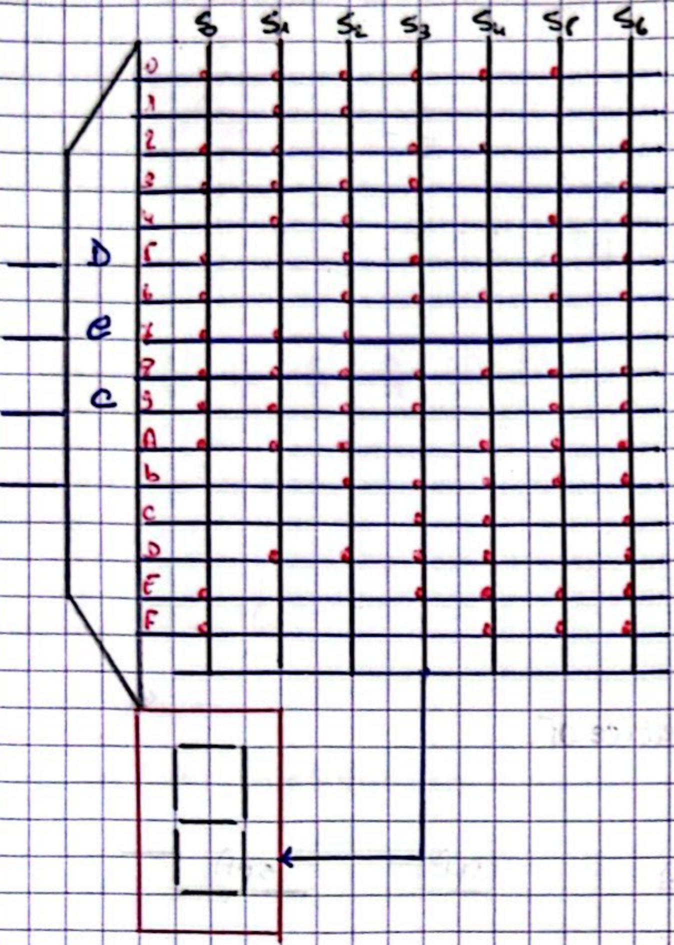
word @		Block @

Block selection (have the same Block @)

05 . 11 . 2023

Computer Memory: ROM

Exercice 01:



Exercice 02:

1)

Q_3	Q_2	Q_1	Q_0	Q_3^+	Q_2^+	Q_1^+	Q_0^+
0	0	0	0	0	0	0	1
0	0	0	1	0	0	1	0
0	0	1	0	0	0	1	1
0	0	1	1	0	1	0	0
0	1	0	0	0	1	0	1
0	1	0	1	0	1	1	0
0	1	1	0	0	1	1	1
0	1	1	1	1	0	0	0
1	0	0	0	1	0	0	1
1	0	0	1	1	0	1	0
1	0	1	0	1	0	1	1
1	0	1	1	1	1	0	0
1	1	0	0	1	1	0	1
1	1	0	1	1	1	1	0
1	1	1	0	1	1	1	1
1	1	1	1	0	0	0	0

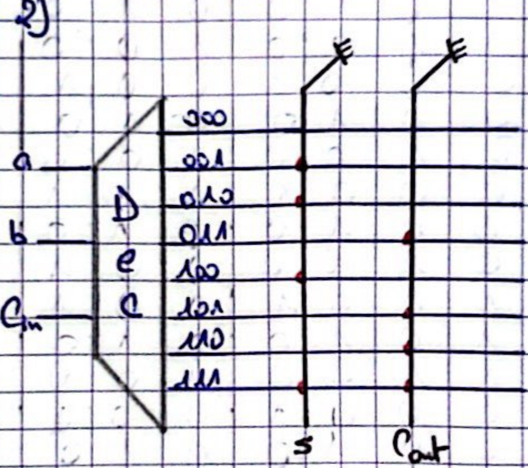
Q + 1

Exercice 01:

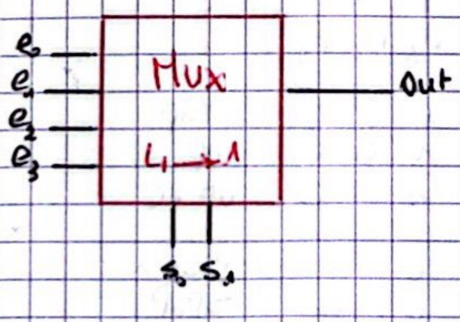
1)

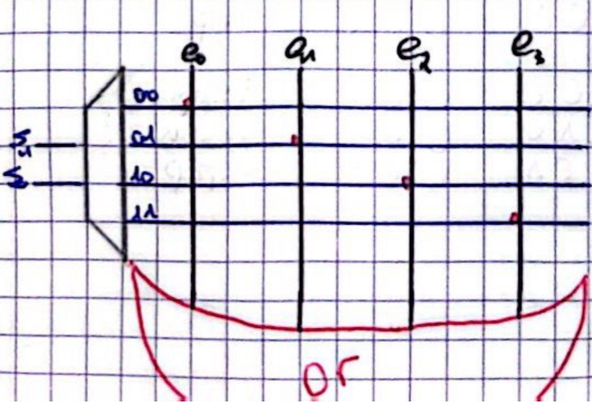
a	b	C_{in}	S	C_{out}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

2)

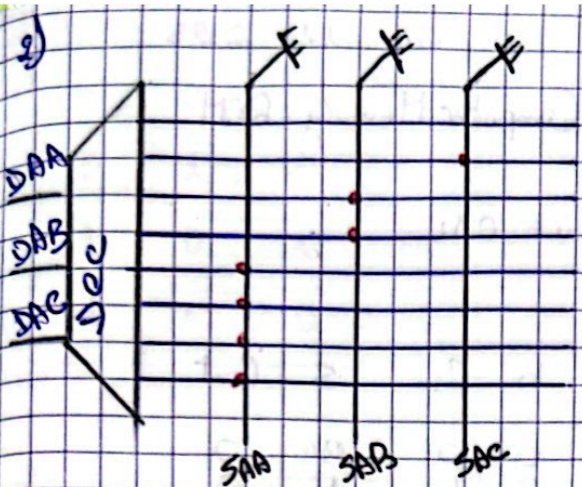


Exercice 03:





2)



Exercice 06 :

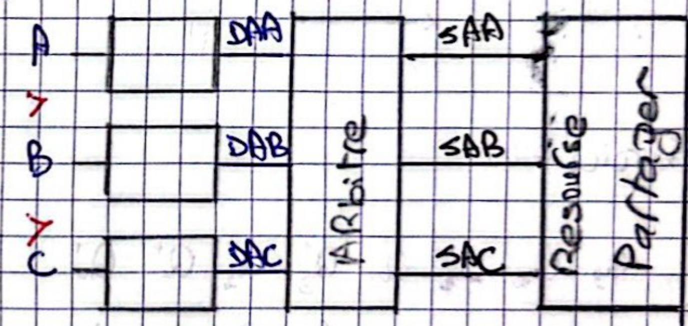
1)

senses			inputs			outputs		
F_3	F_2	F_1	UU	US	UD	UP_0	$STOP_0$	$Down_0$
0	0	1	0	0	1	1	0	0
0	0	1	0	1	0	0	1	0
0	0	1	1	0	0	1	0	0
0	1	0	0	0	1	0	0	1
0	1	0	0	1	0	0	1	0
0	1	0	1	0	0	1	0	0
1	0	0	0	0	1	0	0	1
1	0	0	0	1	0	0	1	0
1	0	0	1	0	0	0	0	1
0	1	1	0	0	1	0	0	1
0	1	1	0	1	0	0	1	0
0	1	1	1	0	0	1	0	0
1	1	0	0	0	1	0	0	1
1	1	0	0	1	0	0	1	0
1	1	0	1	0	0	1	0	0
x	x	x	0	0	0	0	0	0

user	senses	
F_1 F_2 F_3	F_1 F_2 F_3	
1 0 0	1 0 0	STOP
0 1 0	1 0 0	UP
0 0 1	1 0 0	UP+C
1 0 0	0 1 0	Down
0 1 0	0 1 0	STOP
0 0 1	0 1 0	UP
1 0 0	0 0 1	Down+C
0 1 0	0 0 1	Down
0 0 1	0 0 1	STOP

Exercice 05 :

1)

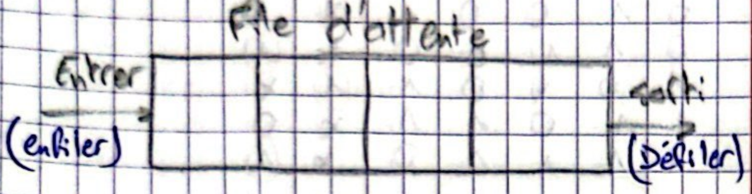


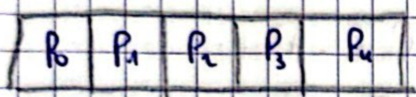
DA : Demande d'accès
SA : signal d'autorisation.

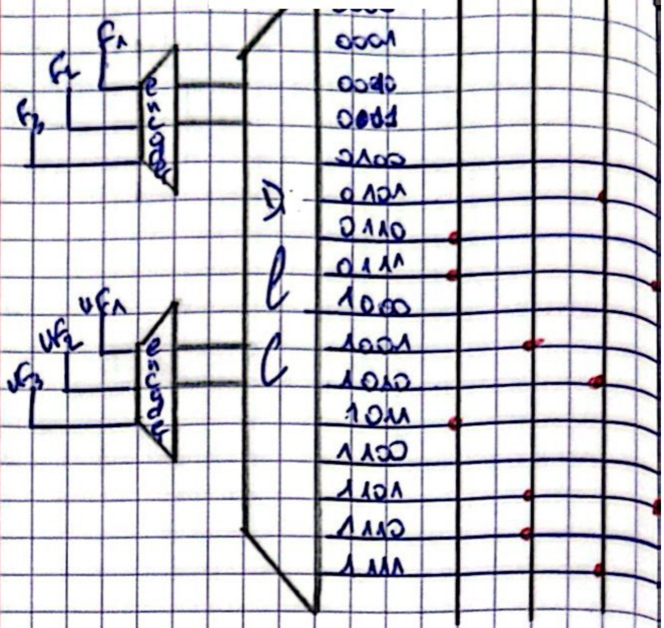
DAA	DAB	DAC	SAA	SAB	SAC
0	0	0	0	0	0
0	0	1	0	0	1
0	1	0	0	1	0
0	1	1	0	1	0
1	0	0	1	0	0
1	0	1	1	0	0
1	1	0	1	0	0
1	1	1	1	0	0

Computer Memory: FIFO + LIFO

Exercice 01:







Ecriture Possible :

$P_W = \overline{P_0}$

Lecture Possible :

$P_R = P_{n-1}$

Logique de décalage :

P_{i-1}	P_i	P_{i+1}	P_{i-1}^+	P_i^+	P_{i+1}^+	J_i	K_i
0	0	0	0	0	0	0	X
0	0	1	0	0	1	0	X
0	1	0	0	0	1	X	1
0	1	1	0	1	1	X	0
1	0	0	0	1	0	1	X
1	0	1	0	1	1	1	X
1	1	0	1	0	1	X	1
1	1	1	1	1	1	X	0

$J_i = P_{i-1} \cdot \overline{P_i}$

$K_i = P_i \cdot \overline{P_{i+1}}$

Etage d'entrer :

P_0	P_1	DE	P_0^+	P_1^+	J_0	K_0
0	0	0	0	0	0	X
0	0	1	1	0	1	X
0	1	0	0	1	0	X
0	1	1	1	1	1	X
1	0	X	0	1	X	1
1	1	X	1	1	X	0

$J_0 = \overline{P_0} \cdot DE$ $\quad$ $K_0 = P_0 \cdot \overline{P_1}$

Etage de sortie :

P_{n-2}	P_{n-1}	DL	P^+_{n-2}	P^+_{n-1}	J_{n-1}	K_{n-1}
0	0	x	0	0	0	0
0	1	0	0	1	x	0
0	1	1	0	0	x	1
1	0	x	0	1	1	x
1	1	0	1	1	x	0
1	1	1	1	0	x	1

$$J_{n-1} = P_{n-2} \cdot \overline{P_{n-1}}$$

$$K_{n-1} = DL \cdot P_{n-1}$$